Title

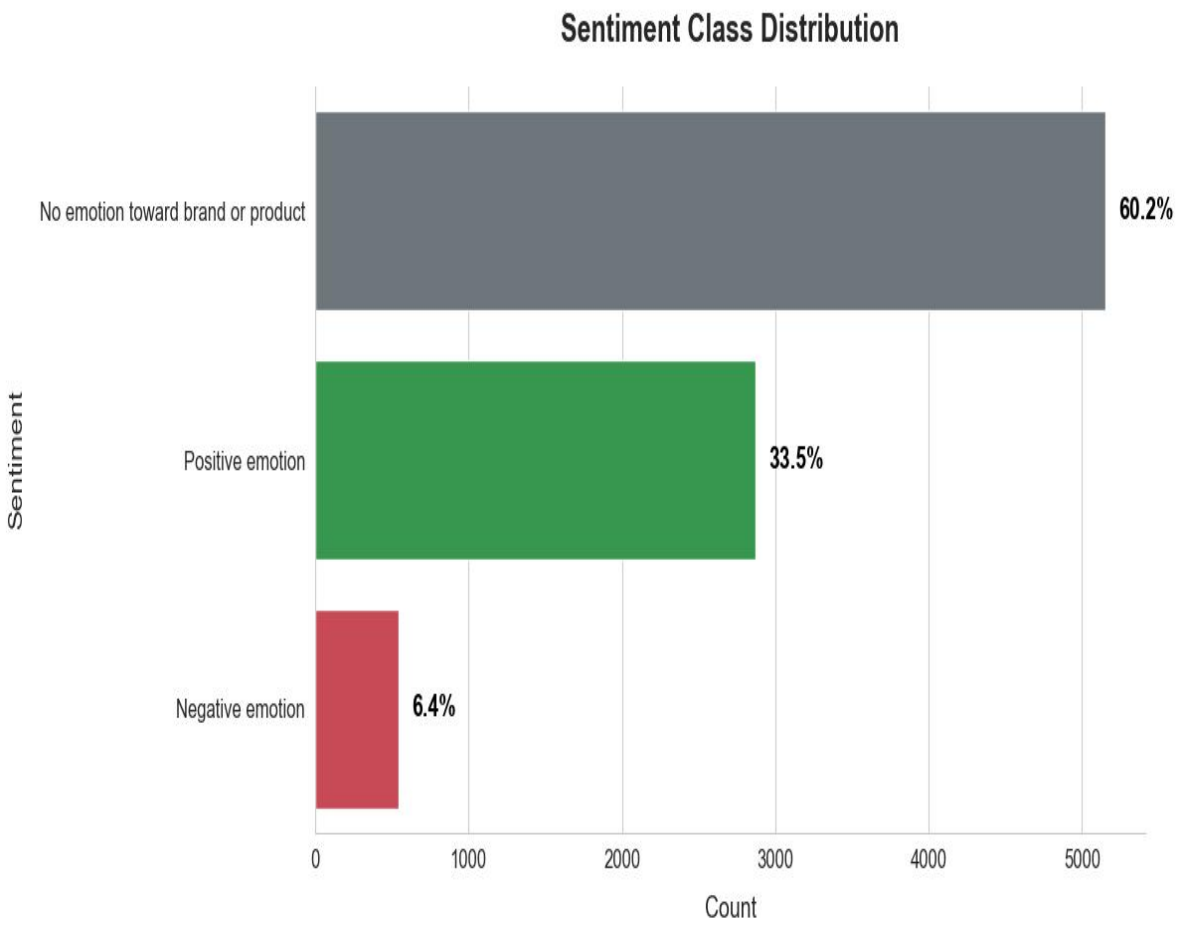
Advanced Sentiment Analysis for Apple & Google SXSW Sentiment Analysis

Introduction

Business & Data Understanding

Primary Audience: Apple & Google Branding Teams

- **Scalability Problem:** Thousands of tweets/hour , manual monitoring impossible
- **Signal vs Noise:**
  - ✓60% = Neutral (noise)
  - ✓6% = Negative (critical insights ⚠)
- **Cost of Delay:**
  - Slow response → brand damage within hours



Business & Data Understanding Cout..

Data + Insight

❖**Dataset:** 8,721 tweets analyzed

❖**Focus:** Apple & Google products

❖**Key Insight :**

✓Negative tweets are rare but critical- creates an **imbalanced classification problem**

Technical Pipeline

Raw Tweets

- ✓ Removed noise (URLs, mentions, hashtags, numbers)
- ✓ Preserved **negations** (e.g.. “not,”, “never”)
- ✓ Applied **lemmatization**

Text Preprocessing

- ✓ Removed noise (URLs, mentions, hashtags, numbers)
- ✓ Preserved **negations** (e.g.. “not,”,“never”)
- ✓ Applied **lemmatization**

Feature Engineering

- ✓ TF-IDF vectorization
- ✓ N-grams (1,2) → words + phrases

Insight:

Negative tweets are rare but critical

Modeling: The Smart Filter

1. What is the Model?

❖AI classification = **high-speed sorting system**

❖Detects **negative sentiment automatically**

❖Filters critical complaints from noise

2. Why It Matters

❖Focus only on **important negative tweets**

❖Ignore majority **neutral noise**

❖Enables **fast response to issues**

3. Strategy & Optimization

❖Tested multiple models → selected **CrisisRecallModel**

❖ Goal: **Never miss complaints**

❖Adjusted threshold:

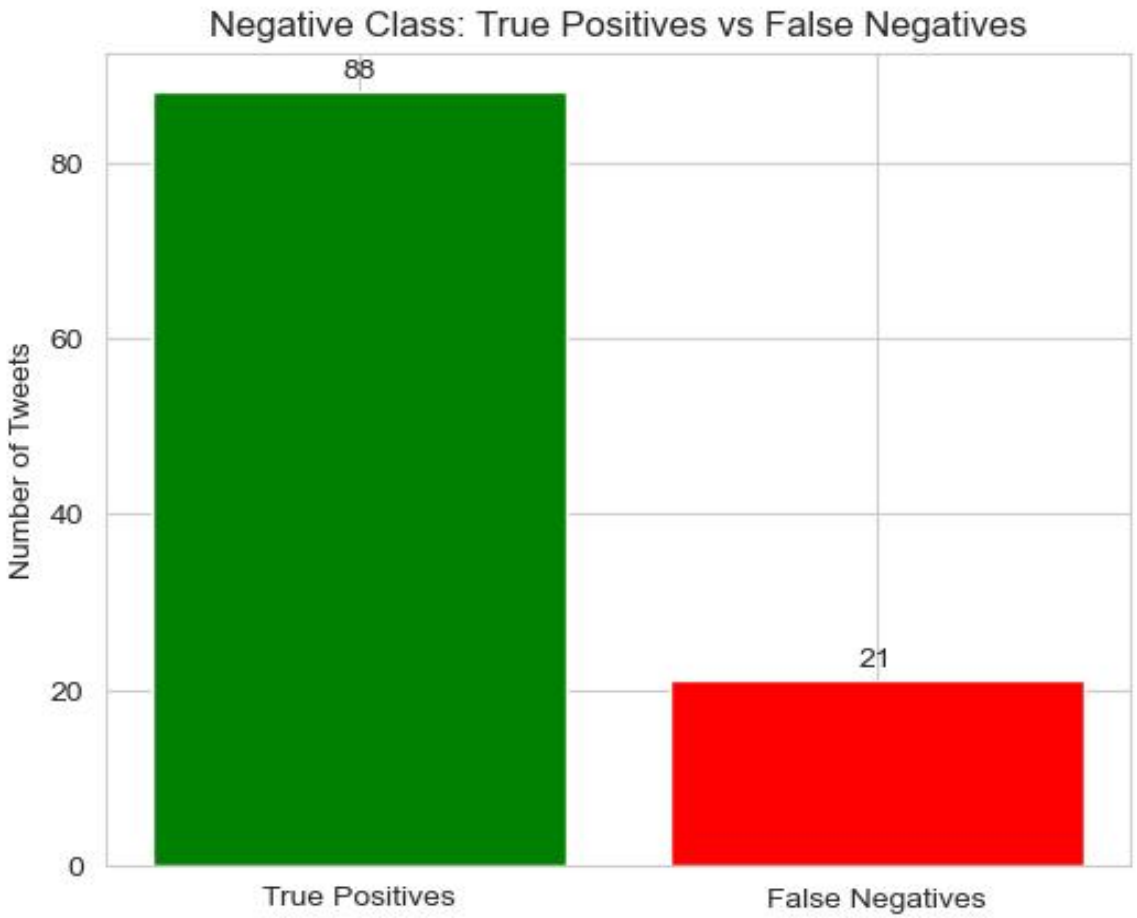
- From **0.50** → **0.15**

4. Effect (Highlight Box):

❖Tweets with ≥15% negative probability are flagged early

Model Comparison Summary

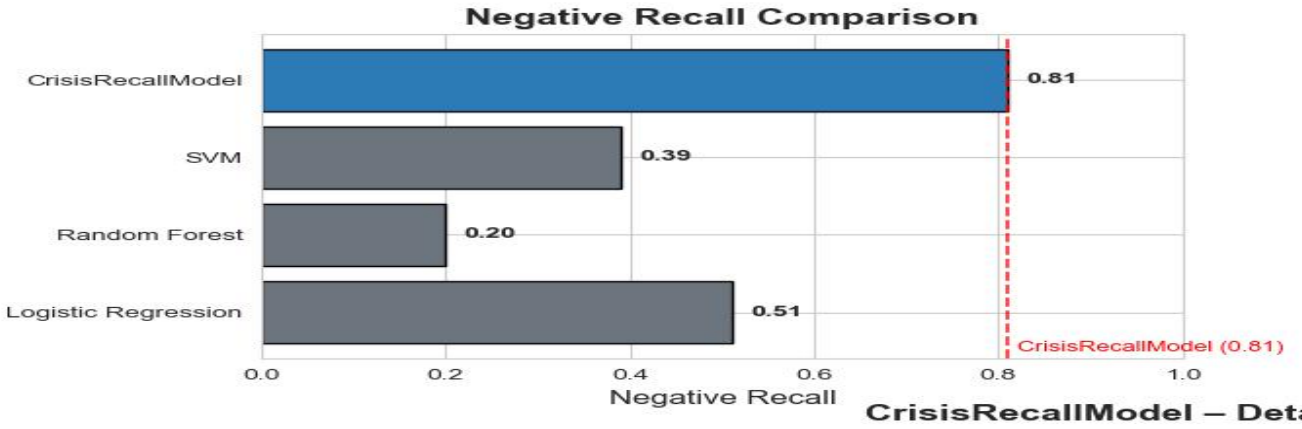
- **Recall improved significantly**
  From 51% → 81% after tuning sensitivity
  We now catch 81 out of every 100 negative tweets automatically
- **Acceptable trade-off**
  Slight increase in false positives (neutral tweets flagged as negative).
  Better to review a few extra tweets than miss a major crisis
- **Reliability confirmed**
  Consistent performance across diverse datasets.
  Model is ready for real-world deployment



Model Comparison Summary Count…

CrisisRecallModel – Catching the Critical 81%

Negative recall improved from 51% → 81% (vs. other models).
Slight drop in overall accuracy is an acceptable trade□off to avoid missing crises.



Model Performance Comparison

Model	Neg Recall	Accuracy
Logistic Regression	0.51	0.63
Random Forest	0.2	0.67
SVM	0.39	0.68
CrisisRecallModel	0.81	0.47

CrisisRecallModel – Detailed Classification Report

Class	Precision	Recall	F1-score	Support
Negative emotion	0.13	0.81	0.23	109.0
No emotion toward brand or product	0.78	0.49	0.6	1031.0
Positive emotion	0.54	0.39	0.45	574.0

Note: Lower overall accuracy is acceptable – priority is catching true negative complaints.

Final Model Performance

Tuned Logistic Regression Results

Negative Recall: 0.8073 (Target Met!).

Overall Accuracy: 47% (A necessary trade-off to ensure high sensitivity to negative feedback).

Key Takeaway

The model now successfully captures over **80%** of all negative tweets, providing a robust early-warning system for stakeholders.

Recommendations & Next Steps

- **Proactive Engagement:** The branding team should use the 81% recall rate to immediately address complaints during events, turning frustrated users into brand advocates through quick resolution
- **Identify Trends:** The model highlights recurring themes in complaints (e.g., "battery life" or "app crashes"), providing the product team with direct feedback on where to focus improvements
- **Phase 2 – Continuous Growth:**
  - **Advanced AI (BERT):** We plan to implement deeper "context-aware" models to understand sarcasm and more complex language
  - **Active Learning:** Create a feedback loop where the branding team can "tell" the model when it makes a mistake, making it smarter every day

Summary:

We have moved from manual monitoring to an automated, high-sensitivity system that protects the Apple and Google brands by **surfacing** 81% of public complaints in real-time.

Questions & Discussion

Thank You

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